

Predicting Forwards' Market Value in Top 5 European Football Leagues Using Formal Metrics

Abstract

This report presents an attempt to build a machine learning model for predicting the market value of a forward in football (sport) based on formal metrics, such as the number of goals scored. The dataset used for this project was scraped from Transfermarkt.com, before undergoing several processing stages to maximize the training & testing process's results. Two supervised learning methodologies were employed for this purpose, as the objective is to output a continuous value as close to a player's actual market value as possible. These methodologies, which are linear regression and random forest regression, were specifically picked to explore different facets in the relationship between the features.

Keywords: *Linear Regression, Random Forest Regression, Supervised Learning, Feature Engineering, Hyperparameter Tuning, Cross Validation, Football, Market Value*

I. Introduction

More lucrative transfers in the recent years have generally involved forwards, such as Kylian Mbappé (180 mil. Euros, 2018), and Harry Kane (100 mil. Euros, 2023). There exists a myriad of factors behind players' market value, such as their brand value and their club's popularity. Having a settled market value for players allow players and clubs to determine their transfer plans. This leads to an interest in understanding how the market value of forwards is evaluated, especially those who belong in a top 5 European football league, which respectively consists of the Premier League (England), LaLiga (Spain), Bundesliga (Germany), Serie A (Italy) and Ligue 1 (France). These leagues were picked due to their popularity, competitiveness and profitability, resulting in a substantial rise in the market value of players who are part of those leagues.

Forwards uniquely were picked for this study, as it is generally more straightforward to measure their performance via e.g. goals and assists, compared to midfielders or defenders, whose performance metrics are dependent on contextual factors such as a team's playing style. To evaluate market values, Transfermarkt.com was selected as the data source. This is a website that records data about footballers, and calculates their market values based on various factors such as performance metrics, and thorough discussions within the Transfermarkt community.

In this project, it is hypothesized that formal metrics (goals, appearances, etc.) directly relate to an increased market value, and could suffice to predict market value close to the actual market value of a forward. There will be attempts to conduct a study to understand the relationship between the features in the dataset to predict the market value of players, before comparing the output to their market value from Transfermarkt.

II. Literature Review

In this report, three literary sources will be reviewed to support the topic study.

The first source was released by He et al. [1]. This source attempted to combine a variety of metrics from WhoScored, The Guardian and Transfermarkt in order to calculate football players' market value. These metrics consisted of performance metrics (goals, assists), playing time (appearances & minutes played) and player ratings and popularity. However, they encountered various issues, such as record linkage and missing data, resulting in difficulties with training the model.

In the second source, Franceschi et al [2]. evaluated Transfermarkt, acknowledging it as an informative indicator of footballers' market value. However, questions were raised over its market evaluation methods, which are mainly based on popular feedback, AKA crowdsourcing. These in turn highlight the dependence of Transfermarkt's market evaluations on subjectivity, prone to either bias or a lack of guaranteed accuracy. It also highlights the importance of contexts surrounding a player, and notes the lack of universal criteria to determine market values. Additionally, in the third source [3], also by Franceschi et al, there was an attempt to determine all possible variables that could influence players' market value, notably the age, playing time and performance (goals & assists). Other factors such as player popularity and international status was also highlighted.

The fourth document by Müller et al. [4] supports the previous paper's argument against Transfermarkt. It states that although Transfermarkt remains the most popular football platform that can calculate more appropriate market value for players, the crowdsourcing factor may have unnecessarily muddled the market fee evaluation process. Using multilevel regression models, the authors attempted to outline formal metrics for market value calculation, such as age, goals, assists and previous market value.

The fifth source [5] by Arrul et al. outlined the metrics for determining their model's performance, which are MSE and R-squared, while applying K-Fold cross validation to assess the model's robustness in order to evaluate footballer's market values based on their FIFA 19 statistics.

Based on the reviewed literary sources, there is interest in only using formal data from a single data source, which is Transfermarkt. This is due to the intuitiveness in their usage to train a machine learning model, while avoiding challenges previously encountered by the literary sources such as merging various different data sources, or converting anecdotal or contextual information into numerical data. Furthermore, there is a certain framework available to build the models of this study upon, such as utilizing supervised learning models with metrics to evaluate their performance such as R-squared. This also provides an opportunity to explore whether formal statistics alone from the available dataset suffice to predict the market value of a forward, compared to the approaches of the previous studies that relied on contextual data in addition.

III. Data Management

A. Data Sourcing & Scraping

Transfermarkt, the source of the dataset used for this study, is a website that gathers comprehensive data about football players. Transfermarkt primarily collects formal data, such as the number of goals scored, or minutes played. Initially, there were several datasets of footballers' profile available

on Kaggle, including those that were scraped from Transfermarkt. However, these datasets were lacking rather relevant features, so the final decision was to scrape the data from the website itself. To obtain the dataset from Transfermarkt, the scraping tool used to obtain the data was a modified version of [<https://www.kaggle.com/code/gimunu/transfermarkt-data-scraping>], having undergone several adjustments to obtain the desired features for usage. Figure 1 below shows an extract of the data scraped from Transfermarkt, displaying the most up to date player information in the 2024/2025 season.

	position	name	current league	current club	age	contract length (days)	Market Value (Euro)	ID	24/25 minutes	24/25 goals	24/25 games	24/25 assists
1	Forward	Robert Lewandowski	LaLiga	FC Barcelona	36	365	15000000.0	38253	2790.0	32.0	35.0	3.0
2	Right Winger	Mohamed Salah	Premier League	Liverpool FC	32	0	55000000.0	148455	3130.0	29.0	37.0	20.0
3	Forward	Harry Kane	Bundesliga	Bayern Munich	31	730	90000000.0	132098	2459.0	29.0	31.0	10.0
4	Forward	Erling Haaland	Premier League	Manchester City	24	3287	200000000.0	418560	3077.0	27.0	35.0	3.0
5	Forward	Kylian Mbappé	LaLiga	Real Madrid	26	1461	160000000.0	342229	3043.0	27.0	37.0	3.0
6	Left Winger	Raphinha	LaLiga	FC Barcelona	28	730	80000000.0	411295	3028.0	24.0	37.0	16.0
7	Right Winger	Ousmane Dembélé	Ligue 1	Iris Saint-Germain	27	1096	55000000.0	288230	2009.0	23.0	30.0	6.0
8	Forward	Omar Marmoush	Premier League	Manchester City	26	1461	60000000.0	445939	2479.0	23.0	32.0	14.0
9	Forward	Mateo Retegui	Serie A	Atalanta BC	25	1096	35000000.0	554903	2134.0	23.0	36.0	4.0
10	Forward	Serhou Guirassy	Bundesliga	Borussia Dortmund	28	1096	40000000.0	270541	2621.0	23.0	31.0	6.0

Figure 1: A screenshot of the scraped data from Transfermarkt in .csv format

After the scraping process, the raw dataset required several more stages of filtering out unneeded data, and determining the importance of each feature, which at this point totaled up to 12. First, players with missing information (N/A) in certain features were removed, in order to avoid worsening model performance, and to ensure consistency in the dataset. This resulted in 630 samples, representing 630 active players listed as forwards playing in a Top 5 European league.

B. Data Filtering & Feature Selection

During the initial testing processes, several unimportant features were gradually removed. After several initial training processes to observe the models' performance, a random forest feature importance verification process was used to narrow down the number of features. The figure 2 below shows a chart comparing the importance of each feature.

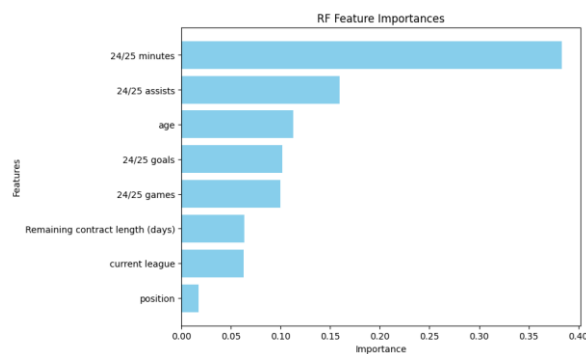


Figure 2: Feature importances in a Random Forest Regression model

Based on Figure 2 above, the number of minutes played seem to play the most important role in determining players' market value. This can be attributed to the fact that players with a larger number of minutes played are usually considered important players. Interestingly, assists are considered more important than the other features, especially goals. This implies that scoring goals are

expected from forwards, but forwards that assist a larger number of goals are deemed to be a positive rarity. Nonetheless, most of the features generally share a similar degree of importance.

Meanwhile, the “position” feature was removed, as initially it was assumed its sub-categories of forwards such as centre forwards and wingers could have allowed a more thorough understanding of the dataset. However, its low importance score indicated that it would have been unnecessary to include this feature in the dataset, alongside other features such as the player's name, their ID number on Transfermarkt and their current club, which are deemed to be too sample-specific.

C. Final Dataset

The finalized features for this study are chosen with the intent of covering different aspects in a forward’s valuation. These aspects are performance (goals & assists), availability (games & minutes played), the top 5 European league they play in, their age, and their remaining contract length. With the exception of the age, higher values in all features equate to higher market value. Figure 3 below shows the format of the final dataset:

	age	current league	Remaining contract length (days)	24/25 games	\	24/25 minutes	24/25 goals/assists
0	36	0.539750	365	35		2790	35
1	32	1.000000	0	37		3130	49
2	31	0.351127	730	31		2459	39
3	24	1.000000	3287	35		3077	30
4	26	0.539750	1461	37		3043	30
...	...	...	...	...		...	...
625	33	0.426682	0	5		61	0
626	21	0.426682	1461	3		45	0
627	29	0.426682	0	14		450	1
628	21	0.351127	365	2		48	0
629	24	0.426682	0	19		459	0

Figure 3: Dataset used to train the models

After extensive testing, the final dataset was left with 6 features, including playing time, goals and assists which were considered important based on the reviewed literary sources. With the dataset finalized, this allowed the opportunity to maximize the models’ performance.

IV. Methodology

This project utilized Python as the programming language, alongside several machine learning libraries such as Numpy for extensive mathematical computations, Scikit-Learn for machine learning algorithms, Matplotlib for data visualization and Pandas for data manipulation. Two supervised learning methods, Linear Regression and Random Forest Regression, were picked due to their different nature. This allows a complementary study of the relationship between the features and the output.

Linear regression is a supervised learning algorithm used to determine a linear relationship between a target variable, and one or more independent variables. When used, it tries to fit a straight line that minimizes the difference between the predicted and actual output values. As it is a straightforward and lightweight method to use, it was ultimately considered for use in this project to explore the linearity between the features.

Random forest regression is a supervised learning algorithm based on decision trees. It consists of training multiple decision trees on a random subset of the data, and a random subset of features. After training, each trees output a prediction, and the model perform a task called aggregation where

all the predictions are averaged into a final prediction. This approach allows the model to reduce variance and overfitting probabilities, leading to more stable predictions. It is often used to handle more complex datasets with high variability, which is applicable in this case as the dataset used come with features of different nature.

V. Testing & Results

A. Testing Metrics

A 10-fold cross validation was applied in order to fully utilize the dataset and reduce variance. Finally, to determine the performance of the models, 3 metrics were considered:

- R^2 , to determine how well the model fits the data.
- RMSE (Root Mean Square Error) to visualize how far the line is fit from the actual value.
- MAE (Mean Absolute Error) to determine the difference between the predicted output and actual output.

B. Initial Model Performance

During the initial testing process, which involved training and testing the model on the dataset as it was, the linear regression model's testing accuracy was close to the training accuracy. The subpar results indicate that the model underfitted and the relationship between the features are not linear. As for the random forest regression, the model overfitted and would require further adjustments.

C. Adjustments to Models & Features

Several measures to improve the models were applied, notably feature engineering for both models and hyperparameter tuning for the random forest regression model. In detail, these measures consisted of:

- Feature transformation:
 - o Applying StandardScaler & log transformation onto the dataset to reduce skewing effects. Improved performance.
 - o Adjusting the presentation of "current league" values from categorical/ordinal values (1, 2, 3, 4, 5) to weighted values using MinMaxScaler. Improved performance.
- Feature selection:
 - o Removed "position", "name", "ID" and "current club". Improved performance.
- Feature creation:
 - o Combined "goals" and "assists" into 1 feature. Improved performance.
 - o Created new features (e.g. minutes per goal). Worsened performance.
- Hyperparameter tuning:
 - o Modified the number of estimators, the maximum depth, the minimum leaf samples and the maximum leaf nodes. Improved performance.

The figure below depicts the evolution of both models' performance before and after gradual adjustments to features and hyperparameters.

Linear Regression - CV Results: R^2 : 0.4411 RMSE: 17,289,991.39 MAE: 10,665,302.96	Linear Regression - CV Results: R^2 : 0.4866 RMSE: 16,570,934.25 MAE: 9,732,238.67
Linear Regression - Training Results: R^2 : 0.4758 RMSE: 16,744,873.80 MAE: 10,414,892.55	Linear Regression - Training Results: R^2 : 0.5197 RMSE: 16,027,903.21 MAE: 9,483,890.87
Random Forest - CV Results: R^2 : 0.5120 RMSE: 16,156,020.33 MAE: 9,509,896.35	Random Forest - CV Results: R^2 : 0.5950 RMSE: 14,718,629.86 MAE: 8,630,227.55
Random Forest - Training Results: R^2 : 0.9310 RMSE: 6,073,955.68 MAE: 3,488,186.35	Random Forest - Training Results: R^2 : 0.8178 RMSE: 9,871,916.49 MAE: 6,449,645.57

Figure 4: A comparison of both models' performance, before and after adjustments

Adjustments resulted in minor improvements, which are approximately 4% for linear regression and 8% for random forest regression for the R-squared score. Nonetheless, the improvement is not considered substantial, especially when the RSME & MAE yielded numbers in the range of millions, a figure that is typically considered non-negligible in the football market. Figure 5 below visualizes the difference between the predicted and the actual market values in both models.

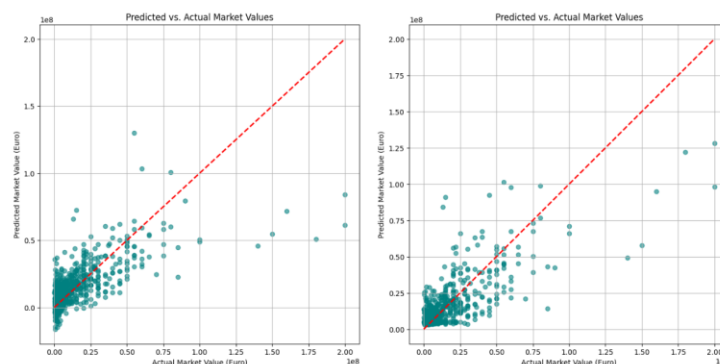


Figure 5: Prediction Accuracy of Linear Regression & Random Forest Regression Models

Based on Figure 5, both models already began to struggle with lower market values between 0 and 50 million euros, often overestimating or underestimating the prediction. As the actual market values began to increase, the models struggled less with overestimations, often undervaluing the market values instead. Meanwhile, the model still relatively managed to correctly predict a considerable number of market values, suggesting that the dataset had relevant information for the model to utilize. This may indicate that additional features for the dataset may be required. The possibility of populating the dataset with more unique features was considered, however several attempts at creating new features based on existing features (e.g. minutes per goal) only further worsened the model's performance. Another considered approach was retrieving player ratings from a separate reputable source, e.g. WhoScored, in order to contextualize the data.

VI. Conclusion

Despite the adjustments, both models respectively underfit and overfit. Although the dataset from Transfermarkt helped narrow down the market value range of forwards in a top 5 European league,

the models were unable to fully analyze the relationship between the features. The overfitting issue concerning the random forest model, which is typically known for being resilient against overfitting, highlights the important presence of nuance in determining market values. It was hypothesized that the dataset required more unique features for the models to fully make use of it, such as the footballers' salary. However, this type of information is rarely made available publicly, furthermore not every footballer has comprehensive data online for use. Nonetheless, there were other features that could have been considered, such as a player's popularity, previous transfer fee and previous market values to determine the trajectory of their market value. Additionally, it could be beneficial to explore in detail how each feature influences the forward market value.

Despite legitimate concerns over Transfermarkt's crowdsourcing aspect in market fee evaluations, such element remains an important factor in determining appropriate market values for players, as it addresses nuances that are not always directly translatable into a numerical format. Should this project be furthered in the future, it would be essential to determine a method to translate contextual information into numeric data and converting them into a standardized format, a challenge previously encountered in the reviewed sources.

VII. References

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